Yash Tajne

Systems Developer | github.com/yashtajne

SUMMARY

Systems developer working in C, C++, Rust, and Go. Started in web development and left it for low-level work: freestanding runtimes, shells, emulators, and desktop tooling. Comfortable with no-libc builds, custom ISAs, and Win32 / Linux / X11 targets.

SKILLS

- Languages: C (freestanding C99), C++, Rust, Go; plus Java, Python, Shell, JavaScript from earlier work.
- Systems: CPU interpreters, custom ISA design, virtual memory, assemblers, MMIO device models, Win32 API, X11.
- Toolchain: Clang / GCC, -nostdlib -ffreestanding builds, batch + shell scripting, Git.

SELECTED PROJECTS (INDEPENDENT)

Pure -- cross-platform C runtime (github.com/yashtajne/Pure)

- Freestanding runtime replacing the libc pieces actually needed: exact-width ints, own bool, size macros, Windows/Linux/macOS detection.
- Relative includes only, no -I flags; Build.py driver. Base layer for PureShell and VIRE.

PureShell -- shell on the Pure CRT (github.com/yashtajne/PureShell)

- Prompt PS <cwd>>, builtins exit / clean / clear / cd, ;-chained commands; parser in Lang/, executor in Program/.
- Links -lPureCore -lPureStd and -lkernel32; builds with build.bat into Shell.exe, zero libc.

VIRE -- virtual hardware emulator, MIT (github.com/yashtajne/VIRE)

- Custom 64-bit ISA (16 GPRs, 3 privilege levels), sparse virtual RAM, assembler; headers in Include/VenV/, impl in Src/.
- Roadmap: console/disk I/O, virtual NIC, then a Linux guest port.

oblivian, Mangobar, cdotenv, todo -- smaller tools (C / Rust)

- oblivian: minimal X11 window manager. Mangobar: status bar in C. cdotenv: dotenv reader in C. todo: small Rust todo.

OPEN SOURCE

25 public repositories at github.com/yashtajne: systems code plus earlier web/ML experiments and Geektoberfest collaborations.

AVAILABILITY

Open to work -- systems / low-level roles (C, C++, Rust, Go). Contact via GitHub: github.com/yashtajne.